

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet**Metallic Bonding — Answers**

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. Attraction between positive metal ions and delocalised electrons.
2. Delocalised electrons.
3. Can be hammered into sheets without breaking.
4. Lattice of positive ions surrounded by a sea of delocalised electrons.
5. Electrons transfer kinetic energy rapidly through the lattice.
6. Layers of ions slide; electrons still hold the structure together.

7. Mg has mobile electrons; MgO ions are fixed in the solid lattice.
8. Different-sized atoms in the alloy hinder layers sliding.
9. Only one delocalised electron per atom / 1+ ions → weaker metallic bonding.
10. Mg^{2+} lattice; delocalised electrons; electrostatic attraction is metallic bonding. [3]

11. Metal layers slide and electrons still attract ions. Ionic lattice: sliding lines up like charges. [4]
12. Delocalised electrons make it a good conductor; it is also ductile. [2]
13. Al forms $3+$ ions and contributes more delocalised electrons than Na ($1+$), so metallic attraction is stronger.